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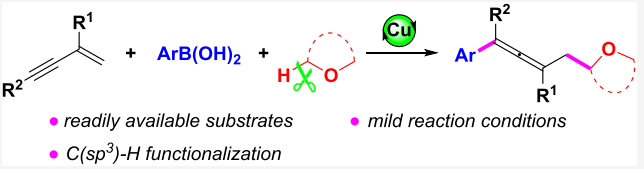
Copper-Catalyzed 1,4-Alkylarylation of 1,3-Enynes with Ethers and Aryl Boronic Acids Enabled via C(sp^3)−H Functionalization

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ABSTRACT: Herein, we present a copper-catalyzed, three-component intermolecular 1,4-alkylarylation of 1,3-enynes with ethers and aryl boronic acids. This method, driven by α-C(sp^3)−H functionalization of the oxygen atom in ethers, regioselectively produces various tetrasubstituted allenes from simple, readily available precursors. Key features include mild reaction conditions and a simple catalytic system.



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INTRODUCTION

Allenes are important structural moieties found in natural products, bioactive molecules, and material sciences, serving as valuable intermediates for diverse transformations in synthetic chemistry.[1] Consequently, numerous efforts have focused on their construction.[2] Among these, 1,4-difunctionalizations of readily accessible 1,3-enynes are a powerful and versatile strategy in organic synthesis to afford multisubstituted allenes.[3] Recently, 1,4-alkylarylation of 1,3-enynes with both alkyl and aryl groups has garnered increasing interest for its ability to diversify allene structures.[4] For example, Hong's group developed a nickel-catalyzed reductive 1,4-alkylarylation method for the catalytic conversion of 1,3-enynes into trisubstituted or tetrasubstituted allenes.[4a] Bao's group reported a copper-catalyzed 1,4-alkylarylation of 1,3-enynes using alkyl diacyl peroxides as masked alkyl electrophiles and aryl boronic acids as nucleophiles.[4b] Among these reactions, direct C(sp^3)−H activation enabling 1,4-difunctionalizations of 1,3-enynes through a radical pathway has gained attention for its ideal atom and step economy in recent years.[5] In this context, Kong and co-workers demonstrated a three-component 1,4-dicarbofunctionalization of 2-trifluoromethyl-1,3-enynes through selective activation of saturated hydrocarbons, combining decatungstate photohydrogen atom transfer with nickel catalysis (Scheme 1a).[5a] Subsequently, Li and co-workers reported a cooperative nickel-photoredox reductive 1,4-dicarbofunctionalization of 1,3-enynes with *N*-methylamines and organohalides, enabled via site-selective C(sp^3)−H functionalization to produce tetrasubstituted allenes (Scheme 1b).[5b] Despite these advances, there remains a high demand for facile and efficient methods to construct multisubstituted allenes from simple, readily available materials.

The construction of carbon−carbon bonds is central to organic chemistry.[6] Therefore, various classical methods have been developed for this purpose. Among these, organoboronic acids are noteworthy as highly effective coupling reagents for constructing C−C bonds owing to their odorless, stable, safe, and easy-to-handle solid form.[7] However, methods employing aromatic boronic acids as coupling reagents for 1,4-difunctionalization of 1,3-enyne via C(sp^3)−H functionalization remain exceedingly rare.[4b,8] In addition, to the best of our knowledge, a copper-catalyzed 1,4-alkylarylation of 1,3-enynes enabled via C(sp^3)−H functionalization has not been reported. In line with our ongoing interests in 1,4-difunctionalizations of 1,3-enynes,[9] we herein report a copper-catalyzed 1,4-alkylarylation of 1,3-enynes with ethers and aryl boronic acids, facilitated via C(sp^3)−H functionalization (Scheme 1c).

RESULTS AND DISCUSSION

We began by exploring a copper-catalyzed 1,4-alkylarylation of 1,3-enynes using oct-1-en-3-yn-2-ylbenzene (**1a**), phenylboronic acid (**2a**), and tetrahydrofuran (THF) as model substrates to optimize the reaction conditions (Table 1; see the Supporting Information for more details). Gratifyingly, employing CuI (5 mol %), 4,4′-di-*tert*-butyl-2,2′-bipyridine (5 mol %), Cs_2CO_3 (1.0 equiv), and *tert*-butyl hydroperoxide (TBHP) (1.0 equiv) under an N_2 atmosphere at room temperature for 2 h produced the desired product **4a** in 28% yield (Table 1, entry 1). The choice of base was critical to this reaction, as using Na_2CO_3 suppressed the desired reaction entirely (Table 1, entry 2). Switching to NaO^{t}Bu improved the yield of **4a** to 45% (Table 1, entry 3). Then, we evaluated

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Scheme 1. 1,4-Alkylarylation of 1,3-Enynes Enabled via C(sp^3)−H Functionalization

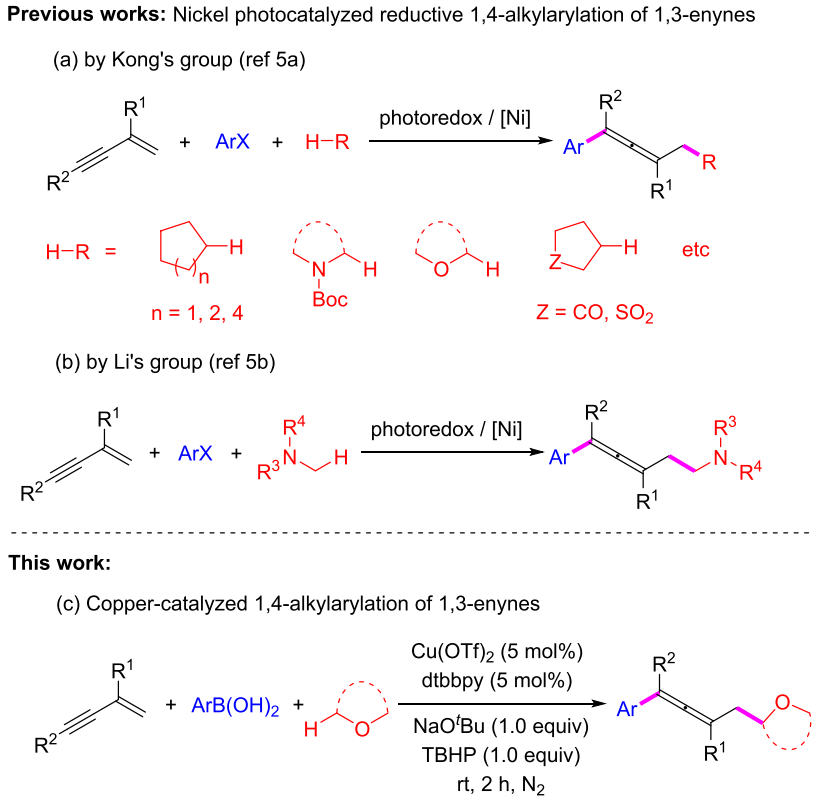
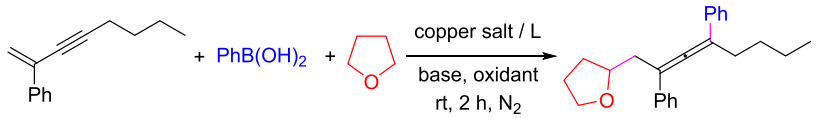


Table 1. Optimization of the Reaction Conditions[a]



1a 2a 3a 4a

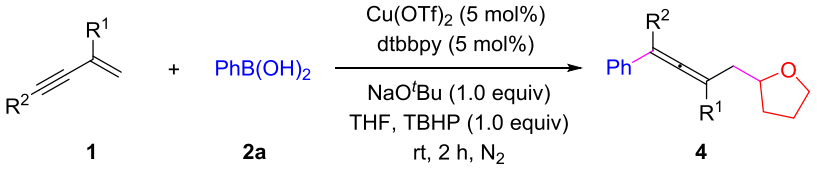
entry	catalyst	ligand	base	oxidant	yield (%)[b]
1	CuI	dtbbpy	Cs_2CO_3	TBHP[c]	28
2	CuI	dtbbpy	Na_2CO_3	TBHP[c]	trace
3	CuI	dtbbpy	NaO^{t}Bu	TBHP[c]	45
4	CuCl	dtbbpy	NaO^{t}Bu	TBHP[c]	7
5	CuOAc	dtbbpy	NaO^{t}Bu	TBHP[c]	39
6	**Cu(OTf)$_2$**	**dtbbpy**	**NaO^{t}Bu**	**TBHP[c]**	**51**
7	Cu(OTf)$_2$	bipy	NaO^{t}Bu	TBHP[c]	43
8	Cu(OTf)$_2$	PPh_3	NaO^{t}Bu	TBHP[c]	0
9	Cu(OTf)$_2$	dtbbpy	NaO^{t}Bu	TBP	trace
10	Cu(OTf)$_2$	dtbbpy	NaO^{t}Bu	$Na_2S_2O_8$	trace
11	Cu(OTf)$_2$	dtbbpy	NaO^{t}Bu	LPO	0
12	none	dtbbpy	NaO^{t}Bu	TBHP[c]	trace
13	Cu(OTf)$_2$	none	NaO^{t}Bu	TBHP[c]	trace
14	Cu(OTf)$_2$	dtbbpy	none	TBHP[c]	trace
15	Cu(OTf)$_2$	dtbbpy	NaO^{t}Bu	none	trace

[a]Reactions were conducted with **1a** (0.3 mmol), **2a** (0.6 mmol), oxidant (0.3 mmol), base (0.3 mmol), catalyst (5 mol %), and ligand (5 mol %) in THF (3.0 mL) under an N_2 atmosphere at room temperature for 2 h unless noted otherwise. [b]Yield of the isolated product. [c]TBHP (5.5 M in decane).

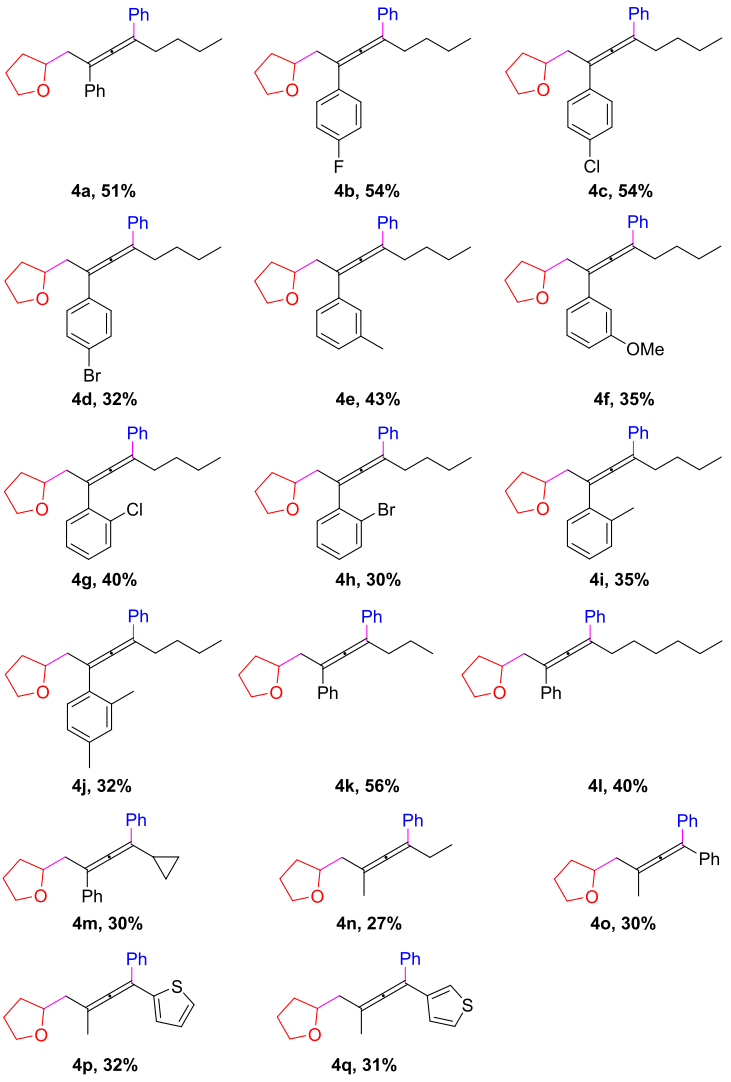
copper catalysts, identifying Cu(OTf)$_2$ as the most effective catalyst (Table 1, entries 4−6). The addition of a ligand was crucial, with bipy yielding 43% of **4a**, whereas PPh_3 showed no activity (Table 1, entries 7 and 8). Further optimization of reaction conditions revealed TBHP as the best oxidant, as alternatives such as di-*tert*-butyl peroxide (TBP), $Na_2S_2O_8$, and dilauroyl peroxide (LPO), were ineffective (Table 1, entries 9−11). Other variations of catalysts, bases, concentration, or reaction temperature resulted in no improvement of the process (see Tables S1 and S4 in the Supporting Information). Control experiments confirmed the necessity of the catalyst, ligand, base, and oxidant, as no desired product was observed

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J. Org. Chem. 2025, 90, 1478−1488

Table 2. Scope of 1,3-Enynes in the 1,4-Alkylarylation Reactions[a]



1 2a 4



[a]Reactions were conducted with **1** (0.3 mmol), **2a** (0.6 mmol), TBHP (5.5 M in decane, 0.3 mmol), NaO^{t}Bu (0.3 mmol), Cu(OTf)$_2$ (5 mol %), and dtbbpy (5 mol %) in THF (3.0 mL) under an N_2 atmosphere at room temperature for 2 h. Yield of the isolated product.

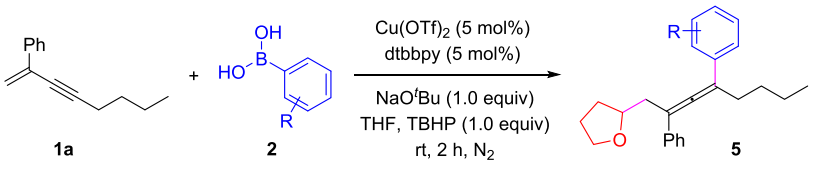
without Cu(OTf)$_2$, dtbbpy, NaO^{t}Bu, or TBHP (Table 1, entries 12−15).

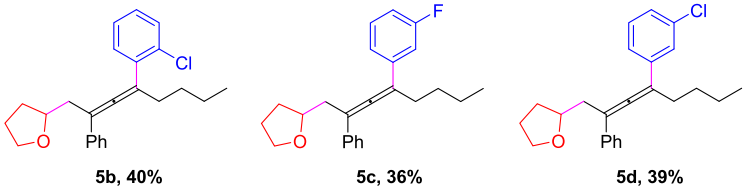
Under the identified optimal reaction conditions (Table 1, entry 6), the substrate compatibility of the current copper-catalyzed 1,4-alkylarylation of 1,3-enynes was investigated. As shown in Table 2, various 1,3-enynes were tested. Substituents on the aromatic ring of the alkenyl moiety, including F, Cl, Br, Me, and OMe at the *ortho*, *meta*, and *para* positions, were tolerated, affording the corresponding products **4b**−**4j** in 30−54% yields. The alkynyl moiety of 1,3-enyne was also investigated, with propyl, *n*-hexyl, or cyclopropyl substitutions providing products **4k**−**4m** in 30−56% yields. Additionally, 1,3-enynes bearing alkyl substituents on both the alkynyl and alkenyl moieties, such as 2-methylhex-1-en-3-yne **1n**, were compatible and afforded the corresponding product **4n**. When R^2 was a phenyl group, as in (3-methylbut-3-en-1-yn-1-yl)benzene **1o**, the reaction produced the desired product **4o** in a 30% yield, probably owing to steric hindrance.[4b] Notably, sulfur-containing heteroaromatic substrates **1p** and **1q** also participated in the reaction, forming the corresponding products **4p** and **4q** in 32% and 31% yields, respectively.

Subsequently, we applied the optimized reaction conditions to a diverse range of aryl boronic acids (Table 3). As expected, various aryl boronic acids bearing different functional groups were well tolerated, yielding the corresponding products in moderate yields. Halo-substituted aryl boronic acids were

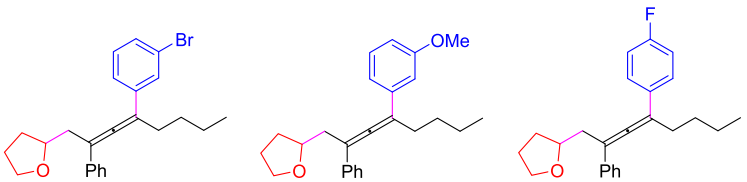
https://doi.org/10.1021/acs.joc.4c02207
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Table 3. Scope of Aryl Boronic Acids in the 1,4-Alkylarylation Reactions[a]

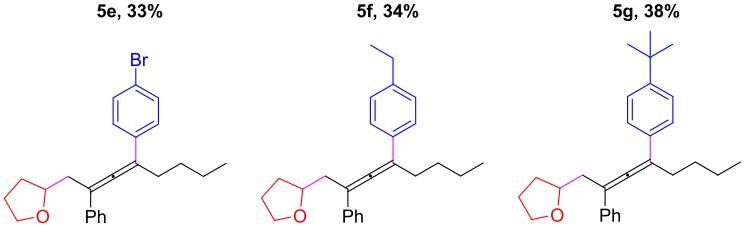




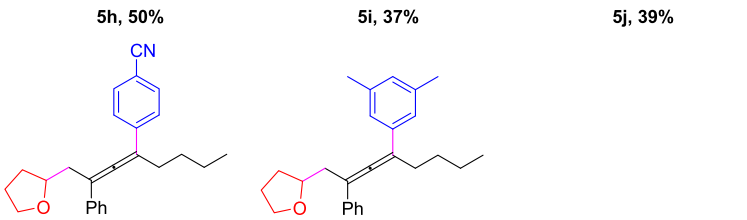
5b, 40% **5c, 36%** **5d, 39%**



5e, 33% **5f, 34%** **5g, 38%**



5h, 50% **5i, 37%** **5j, 39%**



5k, 57% **5l, 45%**

[a]Reactions were conducted with **1a** (0.3 mmol), **2** (0.6 mmol), TBHP (5.5 M in decane, 0.3 mmol), NaO^{t}Bu (0.3 mmol), Cu(OTf)$_2$ (5 mol %), and dtbbpy (5 mol %) in THF (3.0 mL) under an N_2 atmosphere at room temperature for 2 h. Yield of the isolated product.

compatible with this copper-catalyzed 1,4-alkylarylation of 1,3-enynes (**5b**−**5e**, **5g**, and **5h**), indicating their potential for further synthetic transformations. Steric hindrance had a minimal effect on yield, with products **5b** and **5d** isolated in comparable amounts. In addition, substrates bearing disubstituted aryl boronic acids, such as (3,5-dimethylphenyl)-boronic acid, were effective, furnishing product **5l** in 45% yield.

After successfully exploring the scope of 1,3-enynes and aryl boronic acids, we also examined variations of ethers (Table 4). As expected, the reaction proceeded smoothly with similar cyclic ethers, such as tetrahydropyran and 1,4-dioxane, affording the desired products **6b** and **6c** in 50 and 49% yields, respectively. Notably, expanding the scope to acyclic ethers also proved to be successful, yielding the corresponding allenes **6d** and **6e**.

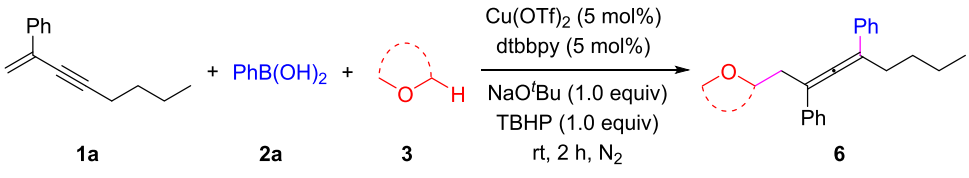
To demonstrate the practical utility of this protocol, a scaled-up reaction was conducted. Satisfyingly, using 1.0 mmol of oct-1-en-3-yn-2-ylbenzene (**1a**) and (4-fluorophenyl)-boronic acid (**2g**) produced **5g** in 34% yield (Scheme 2).

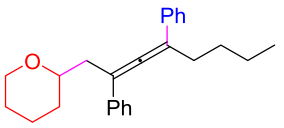
A preliminary mechanistic investigation suggested that the reaction probably proceeded via a radical reaction pathway. This copper-catalyzed 1,4-alkylarylation of 1,3-enynes was completely inhibited when the radical scavenger 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) was introduced (Scheme 3).

Based on the results and related literature, a plausible mechanism is proposed (Scheme 4). Initially, *tert*-butylperoxy (^{t}BuOO$^\bullet$) and *tert*-butoxyl (^{t}BuO$^\bullet$) radicals are generated from TBHP decomposition facilitated by the copper catalyst.[10] Subsequently, these radicals abstract a hydrogen atom from THF, forming ether radical **A**,[11] which reacts with 1,3-enynes to yield propargyl radical **B** and its resonance structure, allenyl radical **C**. Meanwhile, Cu(II) species undergo transmetalation with aryl boronic acid **2a**, affording aryl−Cu(II).[4b,12] Then, aryl−Cu(II) interacts with allenyl radical species **C** to form Cu(III) intermediate **D**. Finally, reductive elimination of the Cu(III) intermediate forms the desired product **4a** and regenerates Cu(I) species,[4b] which are oxidized to Cu(II) to sustain the next catalytic cycle.

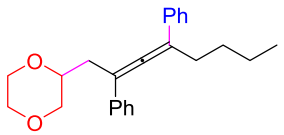
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Table 4. Scope of Ethers in the 1,4-Alkylarylation Reactions[a]

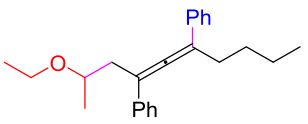




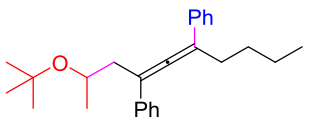
6b, 50%



6c, 49%



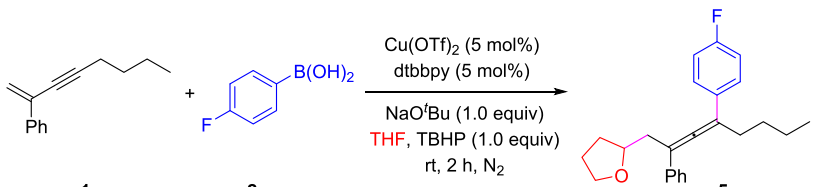
6d, 43%



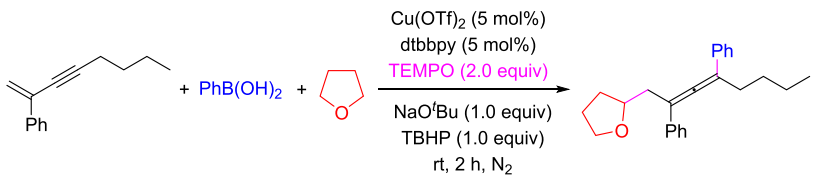
6e, 45%

[a]Reactions were conducted with **1a** (0.3 mmol), **2a** (0.6 mmol), TBHP (5.5 M in decane, 0.3 mmol), NaO^{t}Bu (0.3 mmol), $Cu(OTf)_2$ (5 mol %), and dtbbpy (5 mol %) in ethers (3.0 mL) under an N_2 atmosphere at room temperature for 2 h. Yield of the isolated product.

Scheme 2. Scale-Up Experiments



Scheme 3. Mechanistic Studies



CONCLUSIONS

Herein, we developed a copper-catalyzed three-component 1,4-alkylarylation of 1,3-enynes with ethers and aryl boronic acids via C(sp^3)−H functionalization. This methodology provides a novel method for constructing tetrasubstituted allenes. The developed method is characterized by mild reaction conditions, readily available starting materials, and a simple catalytic system. Ongoing efforts in our group aim to elucidate the detailed reaction mechanism and extend this strategy to the synthesis of multisubstituted allenes.

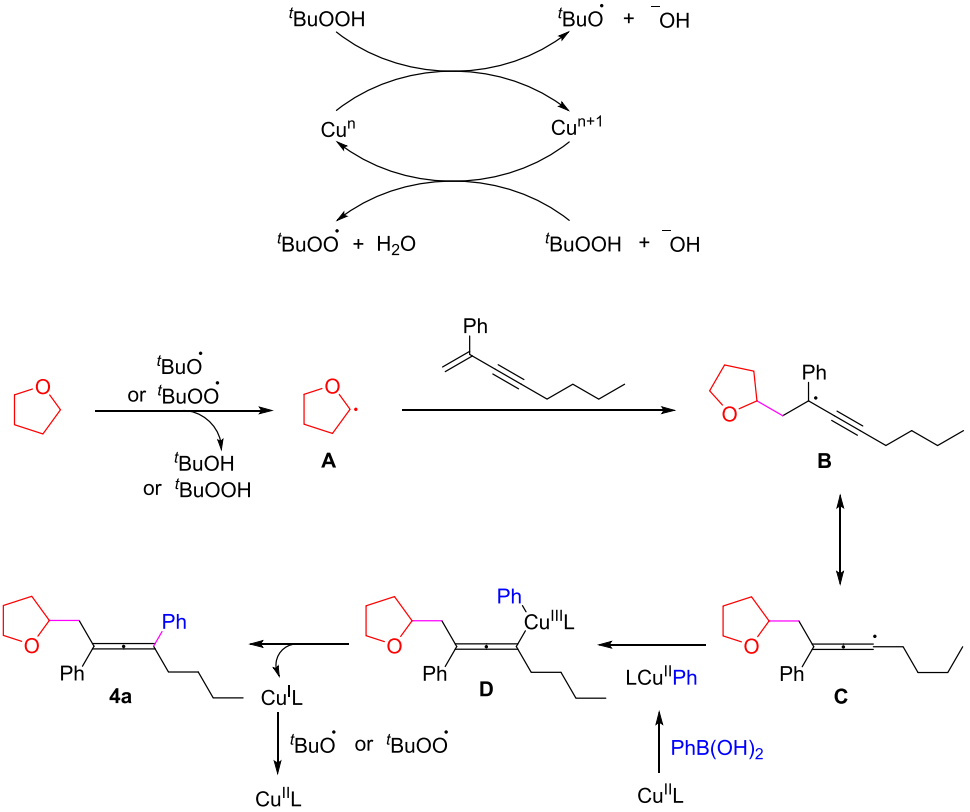
EXPERIMENTAL SECTION

General Information. All deuterated solvents were purchased from Cambridge Isotope Laboratories. ^{1}H NMR, ^{13}C{^{1}H} NMR, and ^{19}F NMR spectra were recorded at 25 °C on a Brüker Advance 400 spectrometer (^{1}H: 400 MHz, ^{13}C: 100 MHz, and ^{19}F: 376 MHz). ^{1}H NMR chemical shifts were referenced to internal $(CH_3)_4Si$ (TMS) at δ 0.00 ppm or to the residual protonated solvent signal: $CDCl_3$ at δ 7.26 ppm. ^{13}C{^{1}H} NMR chemical shifts were referenced to the solvent signal: $CDCl_3$ at δ 77.00 ppm. ^{1}H, ^{13}C, and ^{19}F NMR data were recorded as follows: chemical shift (δ, ppm), multiplicity (s = singlet, d = doublet, t = triplet, m = multiplet, q = quartet, dd = doublet of doublets), coupling constants (Hz), and integration. High-resolution mass spectra were recorded on a Waters LCT Premier-xeTM (USA).

Materials and Methods. Unless otherwise stated, starting materials were purchased from Aldrich or Energy-Chemical Limited and used as received without further purification. Solvents were used directly without further purification. Enynes were synthesized according to literature procedures.[9,13,14] The 1,4-alkylarylation of 1,3-enynes was performed under a nitrogen atmosphere in flame-dried flasks. All reactions were monitored via thin-layer chromatography using Taizhou GF254 silica gel−coated plates. Flash column

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J. Org. Chem. 2025, 90, 1478−1488

Scheme 4. Proposed Mechanism



chromatography was conducted using 200−300 mesh silica gel at increased pressure.

*General Procedure for 1,4-Alkylarylation of 1,3-Enynes. Typical Procedure for Copper-Catalyzed 1,4-Alkylarylation of 1,3-Enynes (**4a** as an Example).* To a solution of oct-1-en-3-yn-2-ylbenzene **1a** (55.2 mg, 0.3 mmol) in tetrahydrofuran (THF, 3.0 mL), phenylboronic acid (74.7 mg, 0.6 mmol), TBHP (5.5 M in decane, 55 μL, 0.3 mmol), NaO^{t}Bu (28.8 mg, 0.3 mmol), Cu(OTf)$_2$ (5.4 mg, 0.015 mmol), and dtbbpy (4.0 mg, 0.015 mmol) were added under a N_2 atmosphere in a Schlenck tube. The reaction mixture was stirred at room temperature for 2 h. After completion, the reaction mixture was purified via column chromatography to yield the corresponding product **4a** (50.8 mg, 51%).

*Scale-Up Experiment for the Synthesis of Product **5g**.* To a solution of oct-1-en-3-yn-2-ylbenzene **1a** (184.1 mg, 1.0 mmol) in tetrahydrofuran (THF, 10.0 mL), (4-fluorophenyl)boronic acid **2g** (285.6 mg, 2.0 mmol), TBHP (5.5 M in decane, 182 μL, 1.0 mmol), NaO^{t}Bu (96.1 mg, 1.0 mmol), Cu(OTf)$_2$ (18.5 mg, 0.05 mmol), and dtbbpy (13.7 mg, 0.05 mmol) were added under a N_2 atmosphere in a Schlenck tube. The reaction mixture was stirred at room temperature for 2 h. After completion, the reaction mixture was purified via column chromatography to yield the corresponding product **5g** (119.5 mg, 34%).

*2-(2,4-Diphenylocta-2,3-dien-1-yl)tetrahydrofuran **4a**.* Colorless oil (50.8 mg, 51%). Petroleum ether/ethyl acetate = 50:1 as an eluent for column chromatography. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.45 (d, J = 7.6 Hz, 4H), 7.31 (t, J = 7.6 Hz, 4H), 7.21 (t, J = 7.6 Hz, 2H), 4.14−4.08 (m, 1H), 3.91−3.86 (m, 1H), 3.72−3.65 (m, 1H), 3.03−2.95 (m, 1H), 2.64−2.54 (m, 3H), 2.04−1.78 (m, 3H), 1.61−1.52 (m, 3H), 1.45−1.39 (m, 2H), 0.91 (t, J = 7.6 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 205.9, 205.8, 136.7, 136.6, 136.6, 136.4, 128.5, 128.4, 126.8, 126.8, 126.8, 126.8, 126.0, 126.0, 125.9, 125.8, 109.3, 109.1, 105.8, 105.7, 77.8, 67.7, 36.8, 36.7, 31.6, 31.6, 30.3, 30.2, 30.2, 30.1, 25.8, 25.8, 22.8, 22.8, 14.0, 14.0. HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{24}H_{28}ONa$ 355.2038; Found 355.2044.

*2-(2-(4-Fluorophenyl)-4-phenylocta-2,3-dien-1-yl)-tetrahydrofuran **4b**.* Colorless oil (56.8 mg, 54%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.45−7.38 (m, 4H), 7.32 (t, J = 7.6 Hz, 2H), 7.21 (t, J = 7.2 Hz, 1H), 6.99 (t, J = 8.4 Hz, 2H), 4.12−4.05 (m, 1H), 3.91−3.86 (m, 1H), 3.72−3.65 (m, 1H), 2.98−2.90 (m, 1H), 2.62−2.53 (m, 3H), 2.05−1.80 (m, 3H), 1.60−1.51 (m, 3H), 1.44−1.39 (m, 2H), 0.91 (t, J = 7.6 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 205.6 (d, J = 2.0 Hz), 205.5 (d, J = 2.0 Hz), 162.9 (d, J = 245.0 Hz), 136.5, 136.3, 132.6, 132.6, 128.5, 127.4 (d, J = 5.0 Hz), 127.3 (d, J = 5.0 Hz), 126.9, 126.9, 126.0, 126.0, 115.4, 115.2, 109.4, 109.2, 105.1, 105.0, 77.8, 77.8, 67.7, 67.7, 37.0, 36.9, 31.7, 31.6, 30.3, 30.2, 30.1, 30.1, 25.8, 25.8, 22.8, 22.8, 14.0, 14.0. ^{19}F NMR (376 MHz, $CDCl_3$): δ −115.96, −115.99. HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{24}H_{27}FONa$ 373.1944; Found 373.1935.

*2-(2-(4-Chlorophenyl)-4-phenylocta-2,3-dien-1-yl)-tetrahydrofuran **4c**.* Colorless oil (59.4 mg, 54%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.45−7.42 (m, 2H), 7.38−7.27 (m, 5H), 7.26−7.20 (m, 2H), 4.11−4.05 (m, 1H), 3.90−3.85 (m, 1H), 3.72−3.65 (m, 1H), 2.96−2.89 (m, 1H), 2.62−2.53 (m, 3H), 2.04−1.78 (m, 3H), 1.60−1.49 (m, 3H), 1.44−1.38 (m, 2H), 0.91 (dt, J_1 = 1.2 Hz, J_2 = 7.6 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 205.9, 205.8, 136.4, 136.1, 135.2, 132.5, 128.6, 128.5, 127.1, 127.1, 127.0, 126.9, 126.0, 126.0, 109.6, 109.5, 105.2, 105.1, 77.8, 77.7, 67.8, 67.7, 36.8, 36.7, 31.6, 31.6, 30.2, 30.2, 30.1, 30.1, 25.8, 25.8, 22.8, 22.7, 14.0. HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{24}H_{27}ClONa$ 389.1648; Found 389.1643.

*2-(2-(4-Bromophenyl)-4-phenylocta-2,3-dien-1-yl)-tetrahydrofuran **4d**.* Colorless oil (39.5 mg, 32%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.44−7.41 (m, 4H), 7.34−7.29 (m, 4H), 7.22 (t, J = 7.2 Hz, 1H), 4.11−4.04 (m, 1H), 3.90−3.85 (m, 1H), 3.72−3.65 (m, 1H), 2.96−2.88 (m, 1H), 2.61−2.53 (m, 3H), 2.05−1.78 (m, 3H), 1.60−1.49 (m, 3H), 1.44−1.38 (m, 2H), 0.91 (dt, J_1 = 1.2 Hz, J_2 = 7.2 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ 205.9, 205.8, 136.3, 136.1, 135.7, 131.5, 128.5, 127.5, 127.4, 127.0, 127.0, 126.0, 126.0, 120.6, 120.6, 109.7, 109.5, 105.2, 105.1, 77.8, 77.7, 67.7,

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J. Org. Chem. 2025, 90, 1478−1488

67.7, 36.7, 36.6, 31.6, 31.6, 30.2, 30.2, 30.1, 30.0, 25.8, 25.7, 22.8, 22.7, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}BrONa$ 433.1143, 435.1123; Found 433.1134, 435.1117.

2-(4-Phenyl-2-(m-tolyl)octa-2,3-dien-1-yl)tetrahydrofuran **4e**. Colorless oil (44.7 mg, 43%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.47−7.44 (m, 2H), 7.31 (t, J = 7.6 Hz, 2H), 7.26−7.18 (m, 4H), 7.03 (d, J = 7.2 Hz, 1H), 4.12−4.08 (m, 1H), 3.91−3.86 (m, 1H), 3.72−3.65 (m, 1H), 3.04−2.95 (m, 1H), 2.62−2.53 (m, 3H), 2.33 (s, 3H), 2.02−1.79 (m, 3H), 1.59−1.54 (m, 3H), 1.45−1.39 (m, 2H), 0.91 (dt, J_1 = 2.0 Hz, J_2 = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.8, 205.7, 138.0, 136.7, 136.5, 136.5, 128.4, 128.3, 127.6, 126.7, 126.7, 126.5, 126.5, 126.0, 126.0, 123.0, 123.0, 109.1, 108.8, 105.8, 105.7, 77.8, 67.7, 67.7, 36.9, 36.8, 31.6, 31.5, 30.3, 30.2, 30.1, 30.1, 25.8, 25.8, 22.8, 22.8, 21.6, 14.0, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{30}ONa$ 369.2194; Found 369.2194.

2-(2-(3-Methoxyphenyl)-4-phenylocta-2,3-dien-1-yl)-tetrahydrofuran **4f**. Colorless oil (38.1 mg, 35%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.45−7.43 (m, 2H), 7.31 (t, J = 7.6 Hz, 2H), 7.25−7.19 (m, 2H), 7.05 (d, J = 8.4 Hz, 1H), 7.01 (t, J = 2.0 Hz, 1H), 6.78−6.76 (m, 1H), 4.12−4.08 (m, 1H), 3.91−3.85 (m, 1H), 3.78 (s, 3H), 3.72−3.66 (m, 1H), 3.01−2.93 (m, 1H), 2.62−2.53 (m, 3H), 1.95−1.76 (m, 3H), 1.63−1.38 (m, 5H), 0.91 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.9, 159.8, 138.3, 136.4, 129.4, 128.4, 126.8, 126.0, 118.4, 112.2, 111.7, 109.3, 105.7, 77.9, 67.7, 55.2, 36.7, 31.6, 30.3, 30.1, 25.8, 22.8, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{30}O_2Na$ 385.2143; Found 385.2141.

2-(2-(2-Chlorophenyl)-4-phenylocta-2,3-dien-1-yl)-tetrahydrofuran **4g**. Colorless oil (44.0 mg, 40%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.51 (t, J = 6.8 Hz, 2H), 7.37−7.30 (m, 4H), 7.21−7.15 (m, 3H), 3.99−3.95 (m, 1H), 3.87−3.82 (m, 1H), 3.72−3.65 (m, 1H), 2.89−2.81 (m, 1H), 2.63−2.57 (m, 1H), 2.49 (t, J = 7.6 Hz, 2H), 2.05−1.80 (m, 3H), 1.63−1.58 (m, 3H), 1.44−1.37 (m, 2H), 0.93 (dt, J_1 = 3.2 Hz, J_2 = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 204.1, 204.0, 137.5, 137.3, 136.8, 136.6, 132.9, 132.8, 130.6, 130.5, 129.8, 128.3, 128.2, 128.2, 126.7, 126.7, 126.6, 126.6, 126.2, 106.8, 106.7, 104.7, 104.6, 77.8, 77.6, 67.6, 39.8, 39.7, 31.4, 30.2, 30.2, 29.9, 29.8, 25.7, 22.7, 22.6, 14.0, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}ClONa$ 389.1648; Found 389.1651.

2-(2-(2-Bromophenyl)-4-phenylocta-2,3-dien-1-yl)-tetrahydrofuran **4h**. Colorless oil (37.0 mg, 30%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.57−7.50 (m, 3H), 7.37−7.30 (m, 3H), 7.27−7.20 (m, 2H), 7.12−7.08 (m, 1H), 4.00−3.95 (m, 1H), 3.88−3.82 (m, 1H), 3.72−3.65 (m, 1H), 2.87−2.79 (m, 1H), 2.64−2.57 (m, 1H), 2.49 (t, J = 7.6 Hz, 2H), 2.04−1.81 (m, 3H), 1.65−1.58 (m, 3H), 1.45−1.38 (m, 2H), 0.94 (dt, J_1 = 2.8 Hz, J_2 = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 203.7, 203.7, 139.6, 139.4, 136.8, 136.7, 133.0, 130.8, 130.7, 128.5, 128.4, 128.3, 127.3, 127.3, 126.7, 126.6, 126.3, 122.8, 122.8, 107.0, 106.9, 106.2, 106.2, 77.7, 77.5, 67.6, 40.0, 39.9, 31.4, 31.3, 30.3, 30.3, 29.9, 29.8, 25.7, 22.7, 22.7, 14.0, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}BrONa$ 433.1143, 435.1123; Found 433.1146, 435.1129.

2-(4-Phenyl-2-(o-tolyl)octa-2,3-dien-1-yl)tetrahydrofuran **4i**. Colorless oil (36.4 mg, 35%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.47 (t, J = 8.0 Hz, 2H), 7.33−7.27 (m, 3H), 7.21−7.15 (m, 4H), 4.02−3.93 (m, 1H), 3.89−3.84 (m, 1H), 3.72−3.65 (m, 1H), 2.86−2.77 (m, 1H), 2.53−2.45 (m, 3H), 2.35−2.33 (m, 3H), 2.01−1.80 (m, 3H), 1.60−1.53 (m, 3H), 1.43−1.37 (m, 2H), 0.92 (dt, J_1 = 1.2 Hz, J_2 = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 203.6, 203.6, 138.0, 137.9, 137.3, 137.1, 135.8, 135.7, 130.4, 128.5, 128.4, 128.3, 128.2, 126.9, 126.9, 126.5, 126.4, 126.3, 125.7, 106.0, 105.8, 105.0, 105.0, 77.7, 77.6, 67.6, 67.6, 40.7, 40.5, 31.5, 30.4, 30.3, 30.2, 30.1, 25.7, 22.7, 22.7, 20.6, 14.0, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{30}ONa$ 369.2194; Found 369.2201.

2-(2-(2,4-Dimethylphenyl)-4-phenylocta-2,3-dien-1-yl)-tetrahydrofuran **4j**. Colorless oil (34.6 mg, 32%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.49−7.45 (m, 2H), 7.32−7.29 (m, 2H), 7.20−7.16 (m, 2H), 6.97 (d, J = 8.0 Hz, 2H), 4.01−3.93 (m, 1H), 3.88−3.83 (m, 1H), 3.72−3.65 (m, 1H), 2.85−2.77 (m, 1H), 2.51−2.44 (m, 3H), 2.31−2.28 (m, 6H), 2.03−1.81 (m, 3H), 1.59−1.54 (m, 3H), 1.43−1.38 (m, 2H), 0.92 (dt, J_1 = 2.4 Hz, J_2 = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 203.7, 203.6, 137.4, 137.2, 136.5, 136.4, 135.7, 135.6, 135.0, 134.8, 131.3, 128.3, 128.2, 128.2, 128.2, 126.4, 126.4, 126.3, 126.2, 105.7, 105.5, 104.9, 104.8, 77.7, 77.6, 67.6, 67.5, 40.7, 40.6, 31.5, 30.4, 30.3, 30.2, 30.1, 25.7, 22.7, 22.7, 21.0, 20.6, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{32}ONa$ 383.2351; Found 383.2356.

2-(2,4-Diphenylhepta-2,3-dien-1-yl)tetrahydrofuran **4k**. Colorless oil (53.5 mg, 56%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.45 (d, J = 7.6 Hz, 4H), 7.33−7.29 (m, 4H), 7.23−7.18 (m, 2H), 4.14−4.08 (m, 1H), 3.91−3.86 (m, 1H), 3.72−3.65 (m, 1H), 3.03−2.95 (m, 1H), 2.64−2.52 (m, 3H), 2.04−1.79 (m, 3H), 1.66−1.52 (m, 3H), 1.00 (d, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.9, 205.8, 136.6, 136.6, 136.6, 136.4, 128.5, 128.4, 126.8, 126.8, 126.8, 126.8, 126.8, 126.8, 126.0, 126.0, 125.9, 125.8, 109.1, 108.9, 105.9, 105.7, 77.8, 67.7, 36.8, 36.7, 32.6, 32.5, 31.6, 31.6, 25.8, 25.8, 21.4, 21.3, 14.3. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{23}H_{26}ONa$ 341.1881; Found 341.1882.

2-(2,4-Diphenyldeca-2,3-dien-1-yl)tetrahydrofuran **4l**. Colorless oil (43.3 mg, 40%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.46 (d, J = 8.0 Hz, 4H), 7.32 (dt, J_1 = 2.0 Hz, J_2 = 7.6 Hz, 4H), 7.21 (t, J = 7.6 Hz, 2H), 4.15−4.08 (m, 1H), 3.92−3.87 (m, 1H), 3.73−3.66 (m, 1H), 3.04−2.96 (m, 1H), 2.64−2.54 (m, 3H), 2.05−1.78 (m, 3H), 1.63−1.51 (m, 3H), 1.41−1.26 (m, 6H), 0.85 (t, J = 6.8 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.9, 205.8, 136.7, 136.6, 136.6, 136.4, 128.5, 128.4, 126.8, 126.8, 126.8, 126.8, 126.8, 126.0, 126.0, 125.9, 125.8, 109.3, 109.1, 105.8, 105.7, 77.9, 67.7, 36.8, 36.7, 31.7, 31.6, 31.6, 30.4, 30.4, 29.4, 29.4, 28.1, 28.0, 25.8, 25.8, 22.7, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{32}ONa$ 383.2351; Found 383.2356.

2-(4-Cyclopropyl-2,4-diphenylbuta-2,3-dien-1-yl)-tetrahydrofuran **4m**. Colorless oil (28.5 mg, 30%). Petroleum ether/ethyl acetate = 50/1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.63−7.60 (m, 2H), 7.41−7.19 (m, 8H), 4.12−4.01 (m, 1H), 3.91−3.85 (m, 1H), 3.71−3.65 (m, 1H), 3.02−2.94 (m, 1H), 2.62−2.54 (m, 1H), 2.08−1.67 (m, 4H), 1.59−1.49 (m, 1H), 0.93−0.83 (m, 2H), 0.64−0.54 (m, 2H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 204.9, 137.0, 136.8, 136.3, 136.2, 128.5, 128.5, 128.4, 127.0, 126.9, 126.2, 126.2, 125.7, 112.3, 112.2, 107.1, 107.1, 77.7, 67.7, 67.7, 36.9, 36.6, 31.6, 31.5, 25.8, 11.4, 11.3, 7.4, 6.9, 6.7, 6.2. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{23}H_{24}ONa$ 339.1725; Found 339.1728.

2-(2-Methyl-4-phenylhexa-2,3-dien-1-yl)tetrahydrofuran **4n**. Colorless oil (19.6 mg, 27%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.41−7.39 (m, 2H), 7.30 (t, J = 7.6 Hz, 2H), 7.17 (t, J = 7.2 Hz, 1H), 4.03−3.96 (m, 1H), 3.89−3.83 (m, 1H), 3.72−3.66 (m, 1H), 2.47−2.39 (m, 3H), 2.19 (dd, J_1 = 7.6 Hz, J_2 = 14.8 Hz, 1H), 2.03−1.79 (m, 6H), 1.56−1.46 (m, 1H), 1.12 (dt, J_1 = 1.6 Hz, J_2 = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 201.7, 201.7, 138.1, 137.9, 128.2, 126.2, 126.1, 125.9, 125.8, 106.5, 106.4, 100.7, 100.6, 78.0, 77.9, 67.6, 40.6, 40.5, 31.5, 31.5, 25.7, 23.2, 23.1, 19.6, 19.6, 12.7, 12.6. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{17}H_{22}ONa$ 265.1568; Found 265.1574.

2-(2-Methyl-4,4-diphenylbuta-2,3-dien-1-yl)tetrahydrofuran **4o**. Colorless oil (26.1 mg, 30%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.37−7.30 (m, 8H), 7.27−7.23 (m, 2H), 4.10−4.03 (m, 1H), 3.89−3.83 (m, 1H), 3.72−3.67 (m, 1H), 2.49 (dd, J_1 = 5.6 Hz, J_2 = 14.8 Hz, 1H), 2.26 (dd, J_1 = 7.6 Hz, J_2 = 14.8 Hz, 1H), 1.93−1.79 (m, 6H), 1.49−1.42 (m, 1H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 203.9,

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J. Org. Chem. 2025, 90, 1478−1488

137.8, 137.7, 128.4, 128.4, 128.2, 128.2, 126.8, 126.8, 108.8, 100.1, 77.8, 67.7, 40.4, 31.4, 25.7, 19.6. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{21}H_{22}ONa$ 313.1568; Found 313.1578.

2-(2-Methyl-4-phenyl-4-(thiophen-2-yl)buta-2,3-dien-1-yl)-tetrahydrofuran ***4p***. Colorless oil (28.5 mg, 32%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.47−7.44 (m, 2H), 7.37−7.33 (m, 2H), 7.30−7.26 (m, 1H), 7.22−7.20 (m, 1H), 6.99−6.97 (m, 1H), 6.93−6.90 (m, 1H), 4.10−4.04 (m, 1H), 3.90−3.83 (m, 1H), 3.73−3.67 (m, 1H), 2.52−2.46 (m, 1H), 2.29−2.23 (m, 1H), 2.01−1.93 (m, 1H), 1.91 (d, J = 3.2 Hz, 3H), 1.89−1.79 (m, 2H), 1.57−1.45 (m, 1H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 203.4, 203.2, 141.8, 141.7, 137.3, 137.3, 128.3, 128.3, 128.2, 128.2, 127.3, 127.2, 125.4, 125.3, 124.5, 103.8, 103.5, 100.9, 100.8, 77.8, 77.7, 67.7, 40.5, 40.4, 31.5, 31.4, 25.8, 25.7, 19.7, 19.4. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{19}H_{20}OSNa$ 319.1133; Found 319.1136.

2-(2-Methyl-4-phenyl-4-(thiophen-3-yl)buta-2,3-dien-1-yl)-tetrahydrofuran ***4q***. Colorless oil (27.2 mg, 31%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.43−7.40 (m, 2H), 7.36−7.32 (m, 2H), 7.30−7.24 (m, 2H), 7.14−7.07 (m, 2H), 4.13−4.02 (m, 1H), 3.89−3.83 (m, 1H), 3.73−3.67 (m, 1H), 2.46 (dt, J_1 = 6.0 Hz, J_2 = 14.4 Hz, 1H), 2.29−2.21 (m, 1H), 1.97−1.91 (m, 1H), 1.90 (d, J = 2.8 Hz, 3H), 1.89−1.78 (m, 2H), 1.51−1.42 (m, 1H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 203.6, 138.6, 137.8, 137.8, 128.3, 128.3, 128.2, 128.2, 128.1, 127.0, 126.9, 125.1, 125.1, 121.8, 121.7, 104.4, 104.3, 99.9, 99.8, 77.8, 77.6, 67.7, 67.7, 40.5, 40.4, 31.4, 31.4, 25.7, 25.7, 19.6, 19.5. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{19}H_{20}OSNa$ 319.1133; Found 319.1137.

2-(4-(2-Chlorophenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5b***. Colorless oil (44.0 mg, 40%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.50 (d, J = 8.0 Hz, 2H), 7.38−7.30 (m, 4H), 7.22−7.16 (m, 3H), 4.19−4.11 (m, 1H), 3.93−3.87 (m, 1H), 3.75−3.69 (m, 1H), 2.95−2.89 (m, 1H), 2.56−2.47 (m, 3H), 2.02−1.81 (m, 3H), 1.57−1.36 (m, 5H), 0.89 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 203.9, 203.6, 137.5, 137.4, 136.7, 136.7, 132.8, 132.8, 130.4, 130.3, 129.9, 129.8, 128.3, 128.2, 128.2, 126.7, 126.7, 126.6, 126.2, 126.1, 107.9, 107.8, 103.7, 103.6, 77.9, 77.9, 67.8, 67.7, 36.5, 36.3, 33.4, 33.4, 31.3, 31.3, 30.2, 30.1, 25.7, 25.6, 22.6, 13.9. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}ClONa$ 389.1648; Found 389.1648.

2-(4-(3-Fluorophenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5c***. Colorless oil (37.8 mg, 36%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.43 (d, J = 8.0 Hz, 2H), 7.32 (t, J = 7.6 Hz, 2H), 7.27−7.14 (m, 4H), 6.91−6.87 (m, 1H), 4.13−4.06 (m, 1H), 3.91−3.86 (m, 1H), 3.73−3.66 (m, 1H), 3.01−2.92 (m, 1H), 2.67−2.50 (m, 3H), 2.04−1.81 (m, 3H), 1.62−1.53 (m, 3H), 1.44−1.39 (m, 2H), 0.91 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 206.0, 206.0, 163.1 (d, J = 243.0 Hz), 163.1 (d, J = 243.0 Hz), 139.4 (d, J = 8.0 Hz), 139.2 (d, J = 7.0 Hz), 136.2, 136.2, 129.7, 129.7, 128.5, 127.00, 127.0, 125.9, 125.9, 121.6 (d, J = 3.0 Hz), 121.6 (d, J = 3.0 Hz), 113.6, 113.4, 112.9 (d, J = 4.0 Hz), 112.7 (d, J = 3.0 Hz), 108.5 (d, J = 12.0 Hz), 108.5 (d, J = 11.0 Hz), 106.4, 106.3, 77.7, 77.7, 67.7, 67.7, 36.7, 36.7, 31.6, 31.6, 30.2, 30.1, 30.0, 25.8, 25.7, 22.7, 22.7, 13.9. ^{19}F NMR (376 MHz, $CDCl_3$): δ −113.41. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}FONa$ 373.1944; Found 373.1945.

2-(4-(3-Chlorophenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5d***. Colorless oil (42.9 mg, 39%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.46−7.42 (m, 3H), 7.32 (t, J = 7.6 Hz, 3H), 7.24−7.16 (m, 3H), 4.13−4.06 (m, 1H), 3.92−3.87 (m, 1H), 3.73−3.66 (m, 1H), 3.01−2.91 (m, 1H), 2.68−2.50 (m, 3H), 2.04−1.81 (m, 3H), 1.59−1.50 (m, 3H), 1.46−1.38 (m, 2H), 0.91 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 206.0, 206.0, 138.9, 138.7, 136.2, 136.1, 134.5, 134.4, 129.5, 128.5, 127.0, 126.7, 126.0, 126.0, 125.9, 125.9, 124.2, 124.1, 108.3, 108.3, 106.4, 106.3, 77.7, 77.6, 67.7, 67.7, 36.7, 31.6, 31.6, 30.1, 30.1, 30.0, 25.8, 25.7, 22.7, 22.7, 13.9. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}ClONa$ 389.1648; Found 389.1647.

2-(4-(3-Bromophenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5e***. Colorless oil (40.7 mg, 33%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.60 (d, J = 9.6 Hz, 1H), 7.43 (d, J = 7.6 Hz, 2H), 7.37 (d, J = 7.6 Hz, 1H), 7.32 (t, J = 7.6 Hz, 3H), 7.24−7.14 (m, 2H), 4.12−4.07 (m, 1H), 3.92−3.87 (m, 1H), 3.73−3.67 (m, 1H), 3.10−2.90 (m, 1H), 2.68−2.49 (m, 3H), 2.04−1.82 (m, 3H), 1.57−1.38 (m, 5H), 0.91 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 206.0, 206.0, 139.2, 139.0, 136.2, 136.1, 129.8, 129.6, 128.9, 128.9, 128.5, 127.0, 125.9, 124.7, 124.6, 122.8, 122.7, 108.2, 106.5, 106.4, 77.7, 77.6, 67.7, 36.7, 31.6, 31.6, 30.1, 30.0, 30.0, 25.8, 25.7, 22.7, 22.7, 13.9. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}BrONa$ 433.1143, 435.1123; Found 433.1144, 435.1128.

2-(4-(3-Methoxyphenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5f***. Colorless oil (36.9 mg, 34%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.44 (d, J = 7.2 Hz, 2H), 7.31 (t, J = 7.6 Hz, 2H), 7.25−7.19 (m, 2H), 7.06−7.03 (m, 2H), 6.77 (dd, J_1 = 2.0 Hz, J_2 = 8.0 Hz, 1H), 4.14−4.07 (m, 1H), 3.91−3.86 (m, 1H), 3.79 (s, 3H), 3.73−3.66 (m, 1H), 3.02−2.94 (m, 1H), 2.64−2.51 (m, 3H), 2.04−1.77 (m, 3H), 1.59−1.51 (m, 3H), 1.44−1.38 (m, 2H), 0.90 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.9, 205.9, 159.7, 138.3, 138.1, 136.6, 129.3, 128.5, 126.8, 125.9, 125.9, 118.6, 118.6, 112.0, 111.9, 109.1, 109.0, 105.9, 105.8, 77.8, 67.7, 55.2, 36.8, 36.7, 31.7, 31.6, 30.3, 30.2, 30.1, 25.8, 22.8, 22.8, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{30}O_2Na$ 385.2143; Found 385.2144.

2-(4-(4-Fluorophenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5g***. Colorless oil (39.9 mg, 38%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.45−7.39 (m, 4H), 7.31 (t, J = 7.6 Hz, 2H), 7.24−7.19 (m, 1H), 6.99 (t, J = 8.4 Hz, 2H), 4.13−4.06 (m, 1H), 3.91−3.86 (m, 1H), 3.72−3.65 (m, 1H), 3.00−2.93 (m, 1H), 2.65−2.51 (m, 3H), 2.03−1.78 (m, 3H), 1.63−1.39 (m, 5H), 0.91 (t, J = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.6 (d, J = 2.0 Hz), 205.5 (d, J = 2.0 Hz), 161.8 (d, J = 244.0 Hz), 136.5, 136.5, 132.6 (d, J = 3.0 Hz), 132.4 (d, J = 3.0 Hz), 128.5, 128.5, 127.5 (d, J = 5.0 Hz), 127.4 (d, J = 5.0 Hz), 126.9, 126.9, 125.8, 125.8, 115.4 (d, J = 1.0 Hz), 115.1 (d, J = 1.0 Hz), 108.4, 108.3, 106.0, 106.0, 77.8, 77.7, 67.7, 67.7, 36.8, 36.7, 31.6, 31.6, 30.3, 30.3, 30.2, 30.1, 25.8, 25.7, 22.8, 22.7, 14.0. ^{19}F NMR (376 MHz, $CDCl_3$): δ −115.95, −115.97. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}FONa$ 373.1944; Found 373.1938.

2-(4-(4-Bromophenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5h***. Colorless oil (61.7 mg, 50%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.42 (d, J = 8.4 Hz, 4H), 7.32−7.30 (m, 4H), 7.22 (t, J = 7.2 Hz, 1H), 4.12−4.05 (m, 1H), 3.91−3.85 (m, 1H), 3.72−3.65 (m, 1H), 2.99−2.92 (m, 1H), 2.65−2.50 (m, 3H), 2.03−1.78 (m, 3H), 1.58−1.38 (m, 5H), 0.91 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.9, 205.8, 136.3, 136.2, 135.7, 135.5, 131.5, 128.5, 128.5, 127.6, 127.6, 127.0, 127.0, 125.9, 125.8, 120.6, 108.5, 108.4, 106.3, 106.3, 77.7, 77.7, 67.7, 67.7, 36.7, 36.7, 31.7, 31.6, 30.2, 30.1, 30.1, 30.0, 25.8, 25.7, 22.7, 22.7, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{27}BrONa$ 433.1143, 435.1123; Found 433.1146, 435.1129.

2-(4-(4-Ethylphenyl)-2-phenylocta-2,3-dien-1-yl)tetrahydrofuran ***5i***. Colorless oil (40.0 mg, 37%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.45 (d, J = 7.6 Hz, 2H), 7.37 (dd, J_1 = 3.2 Hz, J_2 = 8.4 Hz, 2H), 7.30 (t, J = 7.2 Hz, 2H), 7.20 (d, J = 7.2 Hz, 1H), 7.15 (d, J = 8.0 Hz, 2H), 4.14−4.08 (m, 1H), 3.91−3.86 (m, 1H), 3.72−3.65 (m, 1H), 3.03−2.96 (m, 1H), 2.66−2.52 (m, 5H), 2.05−1.80 (m, 3H), 1.60−1.39 (m, 5H), 1.23 (t, J = 7.6 Hz, 3H), 0.90 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.6, 205.6, 142.9, 142.9, 136.8, 136.7, 133.9, 133.6, 128.4, 128.0, 126.7, 126.0, 125.9, 125.9, 125.8, 109.1, 108.9, 105.7, 105.6, 77.9, 67.7, 36.8, 36.7, 31.7, 31.6, 30.3, 30.2, 30.2, 30.1, 28.5, 25.8, 25.8, 22.8, 22.8, 15.5, 15.5, 14.0,

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J. Org. Chem. 2025, 90, 1478−1488

14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{32}ONa$ 383.2351; Found 383.2352.

2-(4-(4-(Tert-butyl)phenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5j***. Colorless oil (45.5 mg, 39%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.44 (d, J = 8.0 Hz, 2H), 7.41−7.28 (m, 6H), 7.19 (t, J = 7.2 Hz, 1H), 4.15−4.08 (m, 1H), 3.92−3.86 (m, 1H), 3.73−3.66 (m, 1H), 3.04−2.96 (m, 1H), 2.63−2.52 (m, 3H), 2.06−1.78 (m, 3H), 1.61−1.54 (m, 3H), 1.44−1.39 (m, 2H), 1.31 (m, 9H), 0.91 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.7, 205.7, 149.8, 149.7, 136.8, 136.8, 133.6, 133.3, 128.4, 126.7, 125.9, 125.8, 125.7, 125.6, 125.4, 109.0, 108.8, 105.7, 105.7, 77.9, 77.9, 67.7, 36.9, 36.7, 34.5, 31.7, 31.6, 31.3, 30.3, 30.2, 30.1, 30.1, 25.8, 22.8, 22.8, 14.0, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{28}H_{36}ONa$ 411.2664; Found 411.2668.

4-(2-Phenyl-1-(tetrahydrofuran-2-yl)octa-2,3-dien-4-yl)-benzonitrile ***5k***. Colorless oil (61.1 mg, 57%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.60−7.53 (m, 4H), 7.41 (d, J = 8.0 Hz, 2H), 7.33 (t, J = 7.6 Hz, 2H), 7.23 (d, J = 7.6 Hz, 1H), 4.11−4.04 (m, 1H), 3.91−3.85 (m, 1H), 3.72−3.65 (m, 1H), 2.97−2.89 (m, 1H), 2.71−2.52 (m, 3H), 2.04−1.82 (m, 3H), 1.60−1.39 (m, 5H), 0.92 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 207.2, 207.2, 141.9, 141.7, 135.7, 135.5, 132.2, 132.2, 128.6, 128.6, 127.3, 127.3, 126.5, 126.5, 125.9, 119.1, 109.9, 108.5, 108.3, 107.1, 107.1, 77.6, 67.7, 67.7, 36.7, 36.6, 31.6, 31.6, 30.1, 30.0, 29.8, 29.7, 25.7, 25.7, 22.7, 22.7, 13.9. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{27}NONa$ 380.1990; Found 380.1987.

2-(4-(3,5-Dimethylphenyl)-2-phenylocta-2,3-dien-1-yl)-tetrahydrofuran ***5l***. Colorless oil (48.7 mg, 45%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.45 (d, J = 8.0 Hz, 2H), 7.31 (t, J = 7.6 Hz, 2H), 7.20 (t, J = 7.6 Hz, 1H), 7.07 (s, 2H), 6.86 (s, 1H), 4.15−4.08 (m, 1H), 3.92−3.87 (m, 1H), 3.73−3.66 (m, 1H), 3.05−2.94 (m, 1H), 2.64−2.51 (m, 3H), 2.30 (s, 6H), 2.04−1.79 (m, 3H), 1.61−1.51 (m, 3H), 1.44−1.38 (m, 2H), 0.90 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.8, 137.9, 137.8, 136.8, 136.7, 136.5, 136.3, 128.6, 128.6, 128.4, 126.7, 125.9, 125.8, 123.8, 123.8, 109.2, 109.1, 105.4, 105.3, 77.9, 77.9, 67.7, 67.7, 36.8, 36.7, 31.6, 31.6, 30.3, 30.2, 30.2, 25.8, 25.8, 22.8, 22.8, 21.4, 14.0, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{32}ONa$ 383.2351; Found 383.2353.

2-(2,4-Diphenylocta-2,3-dien-1-yl)tetrahydro-2H-pyran ***6b***. Colorless oil (51.9 mg, 50%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.48−7.43 (m, 4H), 7.30 (t, J = 7.6 Hz, 4H), 7.19 (t, J = 7.2 Hz, 2H), 3.98−3.85 (m, 1H), 3.55−3.26 (m, 2H), 2.84−2.77 (m, 1H), 2.63−2.53 (m, 3H), 1.73−1.32 (m, 10H), 0.91 (t, J = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 206.0, 206.0, 137.0, 136.8, 136.7, 128.4, 128.4, 128.4, 128.3, 126.6, 126.6, 126.1, 126.0, 125.9, 109.0, 108.9, 105.5, 105.4, 76.6, 76.4, 68.5, 68.4, 37.5, 37.4, 32.0, 32.0, 30.2, 30.2, 30.1, 26.0, 26.0, 23.5, 23.5, 22.8, 22.8, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{30}ONa$ 369.2194; Found 369.2197.

2-(2,4-Diphenylocta-2,3-dien-1-yl)-1,4-dioxane ***6c***. Colorless oil (51.2 mg, 49%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.46−7.41 (m, 4H), 7.33−7.29 (m, 4H), 7.24−7.21 (m, 2H), 3.86−3.59 (m, 6H), 3.39−3.33 (m, 1H), 2.77−2.72 (m, 1H), 2.60−2.52 (m, 3H), 1.62−1.53 (m, 2H), 1.45−1.39 (m, 2H), 0.91 (dt, J_1 = 3.2 Hz, J_2 = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 205.7, 205.6, 136.5, 136.5, 136.3, 136.3, 129.6, 128.5, 128.5, 128.5, 128.4, 126.9, 126.9, 126.9, 126.1, 126.1, 125.8, 120.7, 115.3, 109.7, 109.6, 104.5, 104.4, 74.1, 74.0, 71.3, 66.8, 66.7, 66.4, 33.0, 32.9, 30.2, 30.1, 30.1, 22.8, 22.8, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{28}O_2Na$ 371.1987; Found 371.1977.

(2-Ethoxydeca-4,5-diene-4,6-diyl)dibenzene ***6d***. Colorless oil (43.2 mg, 43%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.46−7.43 (m, 4H), 7.31 (t, J = 7.6 Hz, 4H), 7.20 (t, J = 7.2 Hz, 2H), 3.68−3.29 (m, 3H), 2.92−2.87 (m, 1H), 2.60−2.53 (m, 3H), 1.61−1.55 (m, 2H), 1.44−1.39 (m, 2H), 1.27−1.15 (m, 5H), 1.05−1.01 (m, 1H), 0.91 (dt, J_1 = 2.0 Hz, J_2 = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 206.0, 137.0, 136.9, 136.8, 136.7, 128.4, 128.4, 128.4, 126.7, 126.0, 126.0, 126.0, 125.9, 108.9, 108.7, 106.1, 106.1, 74.0, 63.8, 38.0, 30.3, 30.2, 30.2, 30.1, 22.8, 22.8, 20.2, 15.6, 15.5, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{30}ONa$ 357.2194; Found 357.2198.

(2-(tert-Butoxy)deca-4,5-diene-4,6-diyl)dibenzene ***6e***. Colorless oil (48.9 mg, 45%). Petroleum ether/ethyl acetate = 50:1 as eluent for column chromatography. ^1H NMR (400 MHz, $CDCl_3$): δ 7.45−7.43 (m, 4H), 7.33−7.28 (m, 4H), 7.22−7.18 (m, 2H), 3.88−3.81 (m, 1H), 2.85−2.81 (m, 1H), 2.61−2.54 (m, 3H), 1.62−1.55 (m, 2H), 1.46−1.39 (m, 2H), 1.19−1.13 (m, 12H), 0.91 (dt, J_1 = 2.4 Hz, J_2 = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 206.2, 206.0, 136.9, 136.9, 136.9, 136.6, 128.4, 126.7, 126.1, 126.1, 108.7, 108.5, 106.3, 106.1, 73.6, 73.6, 66.7, 66.6, 40.1, 40.1, 30.6, 30.4, 30.3, 30.2, 28.4, 23.2, 22.8, 22.8, 14.0, 14.0. HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{34}ONa$ 385.2507; Found 385.2512.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c02207.

Optimization of the reaction conditions, ^1H NMR, $^{13}C\{^1H\}$ NMR, and ^{19}F NMR spectra for compounds **4**−**6** (PDF)

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Notes

The authors declare no competing financial interest.

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